HBSM: Hybrid Boundary Sensitivity Mechanisms

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. HBSM tests whether sensitivity in frontier hybrid reasoners can be explained and predicted cheaply. The broad forward-only sensitivity claim is wounded by KL Lens-style forward-sensitivity work, so no novelty or positive result is claimed yet. Current Granite Tiny resource-limited scouts fail B1.

1 Current Gate

The branch must pass sensitivity heterogeneity, no-forward predictor, and mechanism gates. If it overlaps with HORN, it should be folded into a unified hybrid-quantization paper. B1 must first reproduce sensitivity heterogeneity on current hybrid reasoners, B2 must beat no-forward baselines without overfitting depth or size, and B3 must add mechanism beyond HORN's boundary-direction test. The current real trace packet builder/checker covers B1. B2 and B3 now have executable follow-up contracts in `experimental/shared/followup_gate_contracts.py`: they require predictor registry hashes, fixed hyperparameter hashes, train-only selection, held-out Spearman and baseline-margin fields, HORN-alignment signs, noise-off controls, and direction-flip controls. These contracts are not model evidence: no cheap-predictor, no-forward-pass, or mechanistic claim is allowed until real B1 promotes and B2/B3 are filled with train/test or leave-one-model-out predictor splits and matched-noise attention-versus-SSM controls.

2 Mac Artifact Status

A deterministic synthetic B1 real-schema rehearsal now validates the layer-sensitivity packet contract. It emits 720 rows, including 480 primary prompt rows and 240 layer-aligned control rows, and aggregates those primary rows to 40 scoring layers. The packet passes the real checker in non-promoting mode with `SCHEMA_REHEARSAL_NOT_PROMOTABLE_SYNTHETIC_HBSM_B1`. This is synthetic-only: it is not model evidence, not a novelty claim, and not GPU evidence. The evaluator derives measured top-decile membership from aggregated `kl_or_nll_drift`; supplied `top_decile_flag` values are accepted only when they match that measured ranking for every prompt row inside each measured layer. The metadata-only model eligibility packet identifies the current live hybrid targets and architecture hashes, but no corresponding weights are cached repo-locally except Granite Tiny. HBSM now has two Granite Tiny resource-limited forward-sensitivity packets: a one-prompt smoke and a two-prompt prompt-repeat scout. Both are checker-valid, non-promoting B1 failures; there is still no real forward-sensitivity evidence that promotes B1. The two-prompt scout removes the one-prompt boundary-top-decile hint: its boundary top-decile count is zero, Fisher $p = 1.0$, and cheap-predictor Spearman is -0.667.

3 Next Evidence

The next admissible HBSM action is not GPU validation. Either preregister a narrower mechanism hypothesis that explains why the two-prompt B1 scout failed, or fold HBSM into the negative/control appendix of the hybrid-quantization workstream.

4 Admissible Real Packet

The real B1 packet must use fixed boundary flags from the shared architecture maps. Each row must report `model_id`, `layer`, `boundary_flag`, `precision_perturbation`, `kl_or_nll_drift`, `cheap_predictor`, `parameter_count`, `weight_norm`, `top_decile_flag`, `random_top_decile`, `train_test_split`, and `control_type`. The packet must include `config.json`, `raw_rows.jsonl`, `summary.json`, `summary.md`, `decision.md`, `prompt_ids_hash`, `architecture_map_hash`, `trace_plan_hash`, the exact `trace_plan_path` whose file SHA-256 matches that hash, matching `row_count`, and pass `experimental.shared.check_gate_packet --mode real --project hbsm`. The current capture-manifest templates are generated by `experimental.shared.hybrid_trace_capture_manifest`; they must be filled from real forward-sensitivity captures before packet building. Served HF IDs are preserved as `served_model_id` while rows are validated against the canonical architecture-map `model_id`; unfilled template markers are rejected before metric coercion. The checker requires both true and false boundary flags, finite sensitivity and predictor fields, all listed controls on the same `(model_id, layer)` scoring set as the primary rows, and a perturbation-off row whose drift is near zero. It also requires exact measured top-decile cardinality $\lceil 0.10 \cdot \text{scoring layers} \rceil$, where scoring aggregates primary prompt rows to one row per `(model_id, layer)`, agreement between supplied top-decile labels and measured drift ranking on every prompt row, an equally sized random baseline that is not enriched, KL-style and activation/outlier comparator controls, and both train and test split rows unless the packet records an explicit resource limit. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The summary must include primary-row, scoring-layer, and prompt counts; top-decile count; random top-decile count; train/test counts; boundary top-decile enrichment; Fisher p-value; cheap-predictor Spearman correlation; baseline Spearman values; and the cheap-predictor margin over the best baseline. The checker recomputes these fields from `raw_rows.jsonl` and rejects stale or hand-filled `summary.json` entries.

Gate	Promotion threshold
B1 sensitivity	Primary boundary-only rows show top-decile enrichment over non-boundary rows with Fisher $p < 0.05$, while matched random flags do not reproduce the enrichment.
B2 predictor	Spearman $\rho \geq 0.6$ with bootstrap lower bound ≥ 0.3.
B3 mechanism	Matched-noise attention drift exceeds SSM drift and aligns with HORN.

Table 1: Pre-registered HBSM gate thresholds before any standalone claim.

Required baseline	Purpose
Random top-decile flags	Tests enrichment beyond chance.
Layer-index baseline	Controls depth effects.
Parameter-count baseline	Controls layer size.
Weight-norm baseline	Controls simple magnitude predictors.
Boundary-only baseline	Tests whether HORN already explains the effect.
KL-style ranking baseline	Separates HBSM from forward-sensitivity ranking.
Activation/outlier baseline	Separates boundary sensitivity from generic outlier quantization.
Perturbation-off row	Catches hook and scoring bugs.
Train/test split	Reduces cheap-predictor overfit.

Table 2: Controls required before HBSM can claim a cheaper sensitivity predictor.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/hbsm/paper/reviewer_pack.md`. It records that HBSM is not camera-ready as a standalone paper until B1–B3 separate the mechanism from existing sensitivity tools.

6 Novelty Boundary

The closest risk is KL Lens-style forward sensitivity for mixed-precision hybrid models KL Lens (2026). HBSM can survive only if it shows a cheaper no-forward predictor or a boundary mechanism on current hybrid reasoners; otherwise it should be folded into HORN or killed.

References

Jason Kong, Nilesh Prasad Pandey, Flavio Ponzina, and Tajana Rosing. A KL Lens on Quantization: Fast, Forward-Only Sensitivity for Mixed-Precision SSM-Transformer Models. arXiv:2604.13440, 2026. https://arxiv.org/abs/2604.13440.